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Polar ordering of linear rod-like polyamide with different linking structure of nonlinear optical chromophores

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Abstract

We have investigated the correlation between the macroscopic physical properties and the microscopic molecular structure for linear rod-like polyamides with different linking structure of nonlinear optical chromophores by in situ corona poling and optical second harmonic generation. These polymers consist of a wholly aromatic rigid backbone poly(*p*-phenylene terephthalamide), PPTA and nonlinear optical chromophores of various sizes. Though all polymers with different linking structure of chromophores possess similar α- and β-glass transition temperatures, their bulk physical properties greatly depend on the microscopic linking structure. Especially the polyamide F-DCVA with a flexibly linked bulky chromophore exhibits a high internal stress due to a thermal contraction of the rigid polyamide backbone, which at room temperature can overcome the orientational force of the externally applied electric field.
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1. Introduction

Hairly-rod polymers based on a wholly aromatic rigid backbone, polyester and polyamide with flexible side-chains have been studied extensively for the last two decades due to their self-organized mesophase structures having better processability and greater solubility [1–5]. The correlation between the microscopic chemical structure and the macroscopic physical properties for the hairly-rod-like polymers with bulky functionalized side groups such as nonlinear optical (NLO) active chromophores [6], photoconducting groups [7], and other functional groups [8] instead of the alkyl chain without functional group have barely been considered in literature up to now. The molecular design of polymeric functional materials should be related to a macroscopic phase behavior to anticipate the possible physical responses. However, the bulky functionalized side groups complicate the interpretation of their physical properties because of their intricate interactions, such as π–π interaction, steric hindrance, rotational barrier. Additionally, the macroscopic response is much more complicated if the polymer has stiff elements in both the side-chain, e.g. the NLO chromophores, and the main-chain.

We have recently reported the NLO properties of side-chain polymers based on a rigid backbone, modified polyimide [9,10], poly(*p*-phenylene terephthalate), PPT [11–13] and poly(*p*-phenylene terephthalamide), PPTA [14,15]. The poled polymer films with wholly aromatic rigid-rod backbone (e.g. PPT and PPTA) exhibit a unique mountain-like domain structure in contrast to general flexible NLO polymers [11,14]. We have observed a considerably

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different mesophase behavior depending on the linking structure of the chromophores, when similar NLO chromophores were attached as side groups to the same polyester backbone [6]. Moreover, the macroscopic response such as micropatterned domain structures and poling dynamics also depended on the mode of the linking structure and the extent of side groups attached to the orienting polyester backbone [13].

In this paper, we investigate the polar ordering process for a series of the linear rod-like stiff polyamides (PPTA). The in situ corona poling and second harmonic generation (SHG) measurements were performed to obtain a better understanding of the correlation between the microscopic chemical structure and the macroscopic physical properties. The investigated materials had different linking structures between the rigid backbone and the NLO chromophore while keeping the same backbone poly(*p*-phenylene terephthalamide), PPTA of the linear rod-shape. Their macroscopic response strongly depends on small differences in the microscopic linking structure. Especially the polymer F-DCVA with a flexible linking structure of the bulky chromophore is worthy of notice, in which a strong internal thermal stress could overcome the effect of the externally applied electric field.

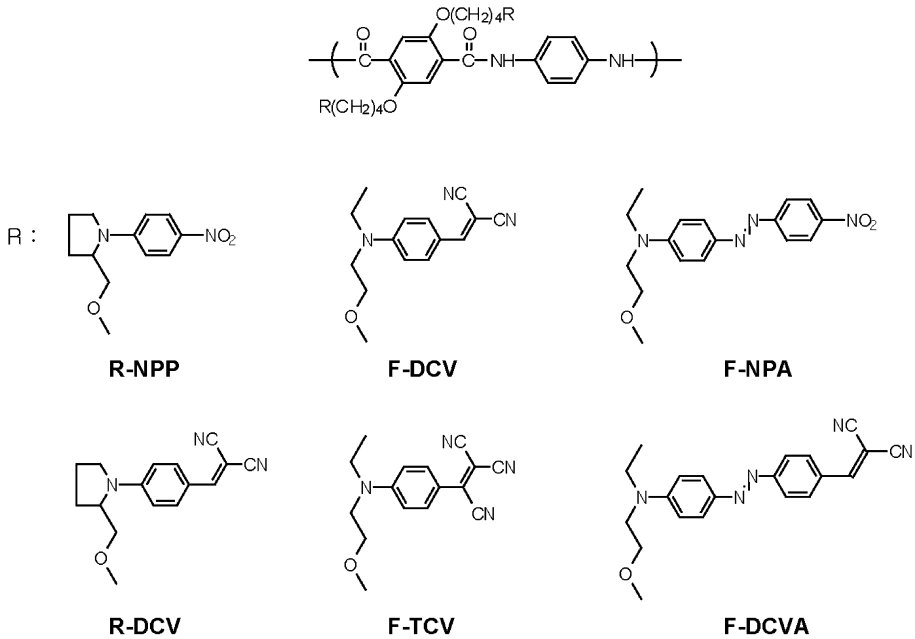
2. Experimental

2.1. Materials

The polymers used here are based on a wholly aromatic rigid backbone polyamide poly(*p*-phenylene terephthalamide), PPTA which has a linear rod-shape resulting from the *para*-position of diacid groups in the terephthalate moiety, and the *para*-position of diamine groups in the *p*-phenylenediamine (see Scheme 1). The NLO chromophores are attached with different linking structures via flexible methylene spacers. The linking structures are either flexible consisting of a conventional amine *N*-ethylanilinoethanol or rigid of (L)-prolinol. As shown in Scheme 1, "F" stands for attachment of the flexible connection and "R" stands for the attachment of the rigid connection. The precursor monomers, 2,5-disubstituted terephthalic acid without NLO functionality were usually postfuctionalized by dicyanovinylation, tricyanovinylation, and azo coupling reaction to form the NLO chromophores. The polymers were prepared by the direct solution condensation of the NLO functionalized terephthalic acid with *p*-phenylenediamine. The details of the syntheses of the monomers and polymers were described elsewhere [14,15]. Note that the alkyl spacer of four methylene units ($n = 4$) was chosen here for the reason that the changes of the length of the alkyl chain resulted in large changes of the NLO response and the polymers with shorter spacers exhibited generally better NLO properties in the previously studied rigid polyester [12].

2.2. Thin film preparation

Polymer films for nonlinear optical measurement were prepared by dissolution of the polymers in *N*-methylpyrrolidine-2-one (NMP), followed by filtration through a 0.45 μm Teflon membrane filter. The solutions were then spin-coated onto indium-tinoxide (ITO) coated glass slides with spinning rate in the range 1000–4000 rpm. After spinning, the samples were dried in vacuum with final pressure



Scheme 1. Chemical structures of the investigated polymers consist of a wholly aromatic rigid backbone poly(*p*-phenylene terephthalamide), PPTA and different linking structure of nonlinear optical chromophores of various sizes.

of less than 0.5 mbar at 50 °C for at least 24 h. The thickness of the films was measured by an Alpha-step profilometer and atomic force microscopy (AFM).

2.3. In situ corona poling and second harmonic generation

The spin-coated films were subjected to a corona discharge using a positive voltage of 5 kV applied to the tip of a tungsten needle placed 10 mm away from the film surface. SHG measurements were made using a Q-switched Nd:YAG laser operating at 1064 nm with a pulse repetition rate of 10 Hz. In order to perform simultaneously the poling and SHG measurement, the p-polarized light was incident at on angle of about 40° to the film surface with an external electric field and various poling temperatures. A reference signal of Y-cut quartz plate was monitored concurrently to reduce laser fluctuations and correct for possible drifts in the laser intensity level. The second-order NLO susceptibilities were measured by a conventional Maker fringe technique [16].

3. Results and discussion

3.1. Thermal properties

The glass transition temperatures of the investigated polymers measured by differential scanning calorimeter (DSC, scan rate 10 K/min) are listed in Table 1. Phase behaviors of rigid backbone polymers with the relatively flexible side-chains are rather complex, while the polymers with flexible backbones usually reveal only one glass transition temperature T_g. In the case of the rigid backbone polymers that we were investigating, the glass transition behavior exhibited two weak and broad endothermic transitions while the motion of side-chains tended to separate from the motion of main-chains [17]. The higher transition temperature is associated with the rigid main-chain motions (α-transition, T_α) and the lower transition temperature is associated with side-chain motions (β-transition, T_β). The glass transition temperature is defined here as the starting point of the broad endothermic transition. These polymers did not show any peaks of thermotropic mesophases, either with the flexible connection or the rigid connection. In general, the glass transition temperature of the functionalized polymers increases with increasing the size and the dipole moment of the side groups [18]. However, the lower and the higher glass transitions in all investigated polymers are both observable as weak broad peaks at approximately similar temperature ranges, around 150 °C for the α-transition characteristic of the polymer backbone and around 85 °C for the β-transition characteristic of the side-chain. Therefore both glass transition temperatures of the highly stiff polyamide of linear rod-shape seem to be independent of the microscopic chemical structure of the side groups such as size, aromaticity, rigidity, and moreover, the linking structure. The glass transition behavior of either the side-chain (β-transition, T_β) or the main-chain (α-transition, T_α) are usually weak and broad for all polymers; in most cases, the DSC thermodiagrams exhibit a changing slope only, without a visible peak and without a clear ending temperature of the transitions. The changes of the heat flow for the glass transitions are more than one order of magnitude smaller than first-order transitions like melting. This is due to complicate interactions of the bulky aromatic side groups with small thermal fluctuations of the stiff backbone. Therefore we could not clearly distinguish the difference in both glass transition temperatures (T_α, T_β) resulting from small differences of the microscopic chemical structures. However, their physical properties, which are presented in the following, show large variations even if the differently linked polymers do not show considerable variations of glass transition temperatures.

3.2. Contribution of rigid main-chain to polar ordering

The spin-coated films were corona-poled under a voltage of 5 kV in air in order to induce the noncentrosymmetric polar order of the NLO chromophores. Then, the nonlinearity was measured using a conventional Maker fringe technique. Since quantitative information on the second-order NLO susceptibility d_{33} values could not be deduced in a similar series of polyesters [12], the effective second-order NLO susceptibility (d_{eff}), defined as a thickness-normalized SHG intensity, was compared for the polymers with different chromophores. It is given by $d_{\mathrm{eff}} = \sqrt{I_{\mathrm{polymer}}/I_{\mathrm{quartz}}}/\ell$, where I is the SHG intensity of light generated by the polymer or the reference quartz plate (1 mm thick), and ℓ is the thickness of the spin-coated film. For the nonlinear optical susceptibility of quartz at 1064 nm a value of $d_{11} = 0.3$ pm/V has been used in accordance with Ref. [19]. The values of thickness-normalized effective NLO susceptibility d_{eff} after poling are listed in Table 1. These results, expectably, showed that polymers

Table 1
General properties of the polymers used in this work: β-glass transition temperature T_β associated with side-chain motions, α-glass transition temperature T_α associated with main-chain motions, and the effective thickness-normalized second-order NLO susceptibility d_{eff} as defined in the text

Polymer	(L)-prolinol (rigid type)		*N*-ethylanilinoethanol (flexible type)			
	R-NPP	R-DCV	F-DCV	F-TCV	F-NPA	F-DCVA
T_β (°C)	80	95	76	79	85	85
T_α (°C)	140	155	145	146	150	145
d_{eff} (μm^{-1})	2.1	2.7	2.9	3.4	5.3	1.7

with chromophores of larger nonlinearity exhibit higher NLO coefficient in the same way like general flexible NLO polymers, except for the F-DCVA polymer which will be discussed later.

Although the series of the investigated polymers show similar glass transition temperatures and follow a general trend of the effective NLO susceptibility d_{eff}, the dynamics of the orientation and relaxation processes strongly depend on the linking structure of the chromophores. Fig. 1 shows the results of SHG experiments by the in situ measurement for two polymers with the same polymer backbone and the same chromophore (i.e. same size and same interaction) with different linking structure. As shown in Fig. 1a, the experimental conditions are the following: first, the samples are heated near to the β-transition temperature T_β (≈100 °C), and then the poling electric field is applied. After stabilization of the initial growth of the SHG signal, the temperature is increased near the α-glass transition temperature T_α (≈170 °C) and subsequently cooled down to room temperature. Comparison of the initial transients shows that increase of the SHG intensity in both samples is similar, but their behavior near the high temperature region related to the α-transition temperature T_α is different: the SHG signal for the R-DCV polymer with the rigid connection increases and stabilizes, while for the F-DCV polymer with the flexible connection it decays when

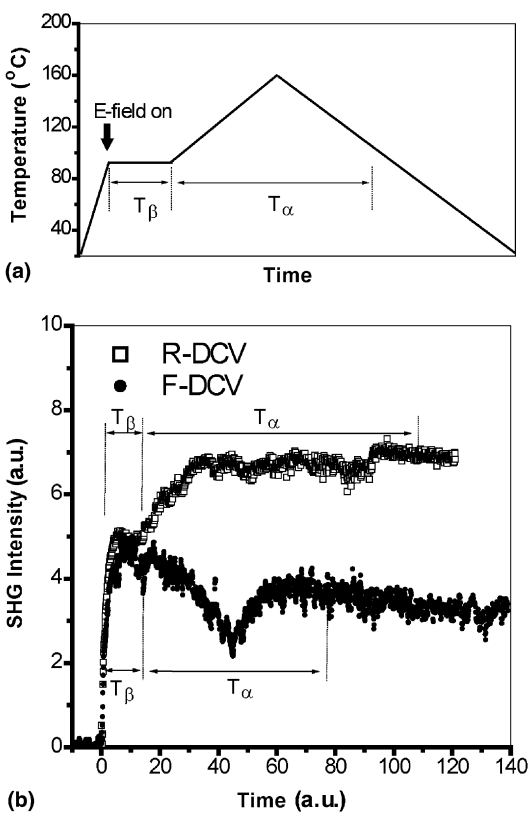


Fig. 1. (a) Poling-temperature sequence. (b) Field-induced growth of the SHG signal for corona poling at 5 kV from the same polymer backbone, PPTA and the same chromophore, DCV with different linking structures for R-DCV (open squares) and F-DCV (filled circles).

increasing the temperature to the α-glass transition region. In the case of the flexibly linked F-DCV polymer the poling field could not overcome the thermal fluctuations of the chromophores in the α-transition temperature region. Therefore, in the case of the F-DCV polymer, the motion of the main-chain with the higher glass transition temperature T_α is not strongly correlated with the motion of the side-chain with the lower glass transition temperature T_β, whereas in the case of the R-DCV polymer the side- and the main-chain motions are strongly correlated. This behavior in the region of the α-glass transition temperature can be explained by the remained mesophase-structure characteristics resulting from the linear linking. Recently, the wholly aromatic polyesters of linear rod-shape have shown a considerably different mesophase behavior [6]: PPT-NPE with a flexible connection exhibited thermotropic mesophases, while PPT-NPP with a rigid connection did not exhibit it. According to the previous results, the flexible connection like in the F-DCV polymer is more likely to result in mesophase characteristics than the rigid connection, even if the mesophase transition cannot be resolved as discussed before. Thus, the side-chain motion in the F-DCV polymer is much more separated from the main-chain motion in contrast to the rigid connection polymer (R-DCV), which exhibits more amophorous characteristics. Therefore, the flexible connection leads to a larger free volume and lower rotational barrier than the rigid connection due to the coupled motion between the main- and side-chains.

However, also in the case of the F-DCV polymer with the flexible linking structure, the side-chain motion is not completely uncoupled from the main-chain motion either, i.e. the rigid main-chain slightly affects the side-chain motion. Fig. 2 shows the relaxation dynamics with different poling conditions. The poling temperatures were 80 °C (near T_β) for the condition A and 150 °C (near T_α) for the condition C, as shown in Fig. 2a. In the condition B, the electric field was turned on at 80 °C (near T_β) like for the condition A, and then the temperature increased to 150 °C (near T_α) like for the condition C. Despite of the large difference of the poling temperatures, similar values of the effective NLO susceptibility $d_{eff} = 2.9 \pm 0.3\ \mu m^{-1}$ were obtained. This is because the alignment of the chromophores in the case of F-DCV is mostly achieved in the lower glass transition temperature T_β as already shown in Fig. 1. In contrast to the steady-state optical nonlinearity, the relaxation for the condition A with only side-chain motion during poling is faster than that of the poling conditions B and C where also a main-chain motion was involved during poling, as seen in Fig. 2b. For the conditions B and C with both side- and main-chain motions a similar relaxation dynamics was obtained. Therefore the motion of the stiff main-chains in the case of F-DCV is still slightly correlated with the side-chain motion.

We can also clearly observe the influence of the remaining mesophase characteristics when comparing the relaxation dynamics for a series of the polymers with different

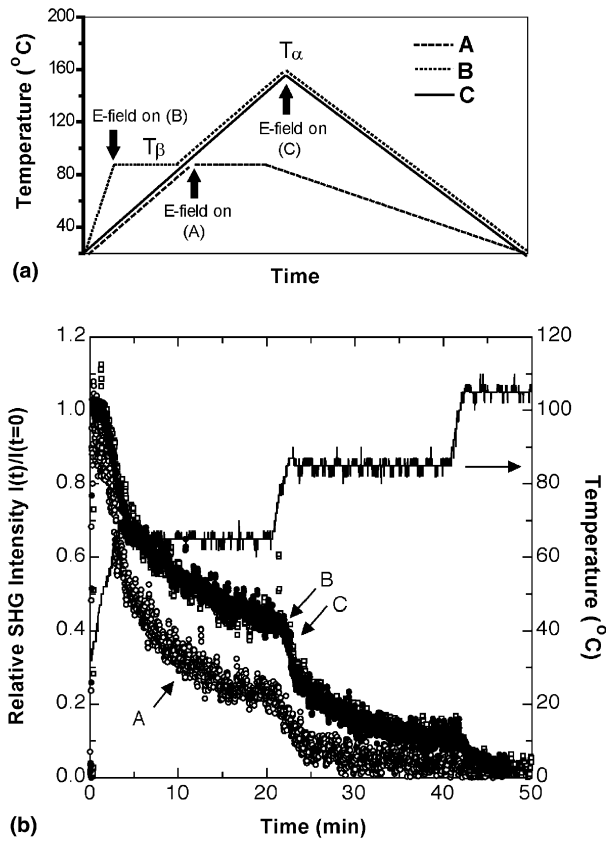


Fig. 2. (a) Poling-temperature sequences for the F-DCV films with treatment conditions A (near T_β, dashed line), B (near T_β and T_α, dotted line), C (near T_α , solid line). (b) SHG signal decay of the polymer F-DCV film with different poling conditions for the conditions A (open circles), B (filled circles), C (open squares).

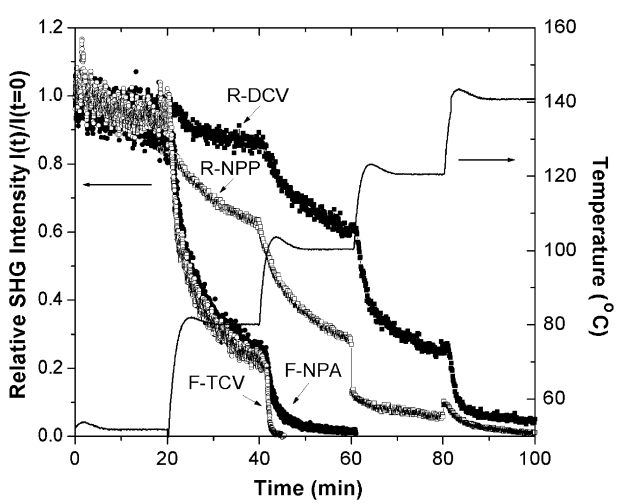


Fig. 3. SHG signal decay of the polymer films for R-NPP (open squares), R-DCV (filled squares), F-TCV (open circles), F-NPA (filled circles). The temperature was stepped to 50, 80, 100, 120, and 140 °C.

chromophores (see Fig. 3). In the case of the polymers with the rigid connection (R-NPP and R-DCV), the temporal stability clearly increases with increasing the size of the chromophores, whereas the polymers with the flexible connection (F-TCV and F-NPA) show a similar dynamics despite of the considerably different size of the chromophores. Moreover, the rigidly connected polymers with smaller chromophores exhibit a more stable orientation of the chromophores than that of the flexibly connected polymers with larger chromophores. This means that the main-chain motion can be helpful to effectively prevent the relaxation of the side-chain motion in the polymers with the rigid connection.

3.3. Internal stress by thermal contraction of linear rigid polyamide

The polar ordering process for the F-DCVA polymer with a flexibly linked chromophore of a large size is unusual as shown in Fig. 4 that shows the development of the SHG signal during poling. The temperature was first increased to the β-glass transition temperature T_β (≈100 °C), and then the electric field was applied. The result clearly shows that the SHG signal rapidly increases and builds up after applying the electric field. However, then the SHG signal unexpectedly decreases to zero during the cooling process. This unexpected behavior also occurs at higher poling temperatures of near or above α-transition temperature and is reproducible after several cycles of the poling process. This behavior implies that certain internal forces of the F-DCVA polymer compensate the influence of the externally applied electric field during the cooling process. Most probably these forces arise due to the internal stress by the thermal contraction of the stiff polyamide main-chains.

In the case of rigid-rod-like polymers, the film prepared by the spin-coating process leads to a mainly planar orientation of the backbones by the flow-induced planar stacking of the rigid segments [1,12,20], and the side groups are tilted with respect to the main-chain layers to decrease

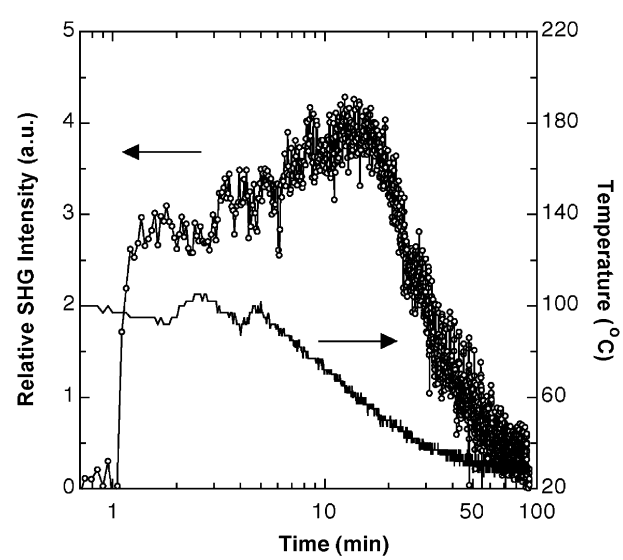


Fig. 4. Corona poling field-induced growth of the SHG signal for F-DCVA polymer. The electric field was applied at time $t_0 = 1$ min.

the distance between the neighboring side groups [5]. Therefore, the chromophores can rotate easily in the plane of the film. The restriction forces in the planar oriented backbones do not usually allow a vertical alignment of the chromophores at room temperature, even with the applied electric field. In the high temperature region the density of the main-chains decreases due to the weaker interactions such as hydrogen bonding, π–π interactions of the main-chains, dipole aggregation force to the chromophores. As in a general NLO polymer, the electric field with elevated temperature in the F-DCVA polymer leads to the parallel arrangement of neighboring chromophores. However, it is still suppressed by the restriction forces mentioned above. Such restricted motion of the bulky chromophores with large microscopic nonlinearities on the F-DCVA polymer lead to macroscopically smaller value of the effective SHG coefficient than that of the polymers with smaller chromophores of lower microscopic nonlinearity (see Table 1).

The reasons for this unexpected phenomena during the cooling process may be related to four effects. First, the distance between the main-chains decreases as the temperature is cooling down. The permanent dipole–dipole interaction varies as r^{-6}, where r is the intermolecular distance. Accordingly, a slight decrease of the interchain distance by the thermal contraction of the planar oriented rigid backbones may induce the compressive force in the direction of the sample normal which is opposite to the direction of the external electric field as illustrated in Fig. 5. Thus, the motion of the chromophores in the interior is more restricted. The internal stress assists to induce the planar orientation of the chromophores in the film and antiparallel dipole–dipole aggregation due to the electrostatic interaction between the chromophores. The second possible reason is related to the rotational barrier of the chromophore. The rotational energy barrier for the rigid connection is approximately two times higher than that of the flexible connection as described in Ref. [6]. This means that the flexible connection is more easily rotated under certain forces. It seems reasonable to expect that the flexible connection is much more favorable to form alignment in the plane compared to the rigid connection. The third suggested reason is the fact that the chromophore DCVA is larger (about 16 Å) than the length of the repeating unit (≈13 Å) [6]. In point of view of the structural barrier related to the size of the chromophores, alignment of DCVA molecules is more difficult than for smaller chromophores. Another possible reason for the unexpected phenomena during the cooling process is that the dipole aggregation progressively increases with increasing the dipole moment caused by more and more electron-withdrawing acceptor groups [21]. Finally this internal stress by the thermal contraction of the main-chains leads a change in the intermolecular configuration from parallel to antiparallel association of the chromophores. Accordingly, in the F-DCVA polymer with flexibly linked chromophores of large size, the orientational force of the externally applied electric field could be suppressed at room temperature.

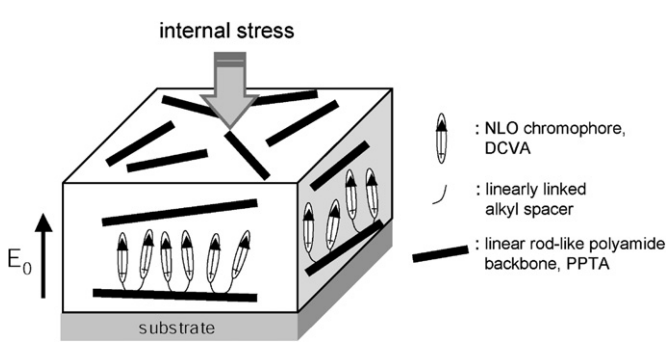


Fig. 5. Schematic diagram of a planar orientation of polymer backbones by flow-induced planar stacking of the rigid segments for F-DCVA polymer, and chromophores oriented normal to the film due to the influence of the external poling electric field.

4. Conclusion

In this study, the correlations between the macroscopic physical behavior and the microscopic molecular chemical structure have been investigated in a series of the linear rod-like polyamide with different linking structure of NLO chromophores by the in situ corona poling and SHG measurements. These polymers consist of a wholly aromatic rigid backbone poly(*p*-phenylene terephthalamide), PPTA and NLO chromophores of various sizes. Though the differently linked polymers show similar α- and β-glass transition temperatures, their physical bulk properties substantially depend on the microscopic structure. Especially in the F-DCVA polyamide with a flexibly linked chromophore of a large size, a high internal stress by a compressive force of the rigid polyamide backbones can overcome the orientational force of the externally induced electric field at room temperature. Future studies on this type of materials will address the observed unexpected phenomenon of the internal stress and also microscopic molecular structure, which should lead to a strong correlation between the main- and side-chain motion in order to achieve a long term temporal stability required for electro-optic devices.

Acknowledgements

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